



Blockchain based Crowdfunding

Munawar Hussain¹, Muhammad Sharif², Fatima Sadiq³, Khair-un-Nisa⁴,

Abstract

This project introduces a blockchain-based crowdfunding system. Traditional platforms are vulnerable to data breaches, high fees, and fraud due to the anonymity of the user's identity. These risks and problems can be solved by blockchain. The blockchain foundation introduces smart contracts to solve these problems. Solidity programming language is used to create smart contracts. In blockchain, there is no central control, making it transparent and a digital identity management system. This project proposes a blockchain-based crowdfunding network that provides privacy, security, and decentralization through the use of Ethereum smart contracts.

Keywords

Blockchain ,Ethereum ,Crowdfunding , Smart contracts

1. Introduction

Crowdfunding is a method of raising funds for a project or venture by soliciting small contributions from a large number of people. Crowdfunding activities have increased worldwide since the onset of the COVID-19 pandemic. It has gained popularity in recent years as it allows individuals to support projects they are passionate about, even if they cannot contribute a significant amount. The primary advantage of crowdfunding is its ability to quickly gather the necessary funds. Blockchain is a decentralized, immutable database that facilitates asset tracking and transaction recording within a corporate network. Solidity and Vyper are programming languages used to develop Ethereum smart contracts. Kickstarter.com, Indiegogo.com, and Mystartr.com are among the most popular crowdfunding platforms.

2. Literature Survey

[1]. In Decentralized Autonomous organization (DAO) based on ethereum. In this system, the registered users can share their drawings, songs, videos or just advertisements. The other users can comment in the post like the post or tip the owner of the post which will cost a certain amount of Likoin; a custom Ethereum ERC-20 token through Initial Coin offerings (ICOs). Each of the above activates gives a certain amount of Likoin to the owner of the post. The likoin tokens can be shared among users.



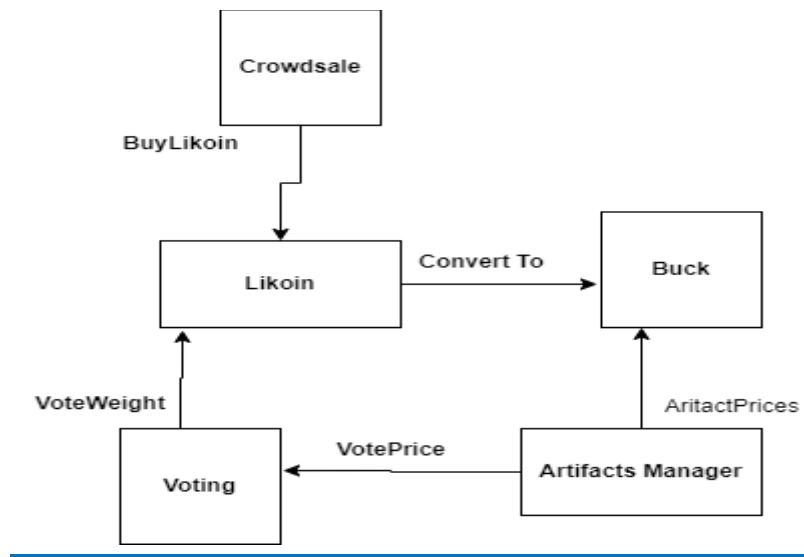


Figure 1: Like Starter smart contracts architecture

[2].In a Modular approach to crowdfunding that allows campaign creators to post their project details or fundraising programs for educational medical or other purposes. This platform utilizes the Ethereum, solidity and a consensus mechanism to ensure safe secure and transparency in transactions. It target both users and organizations, but has the limitation of being difficult for some people to understand due to its dependency on blockchain Technology and the use of the Rinkeby for testing network which was deprecated on October 5,2022. This platform provide unique path to donor and organization to raise funds and bring their ideas to life.

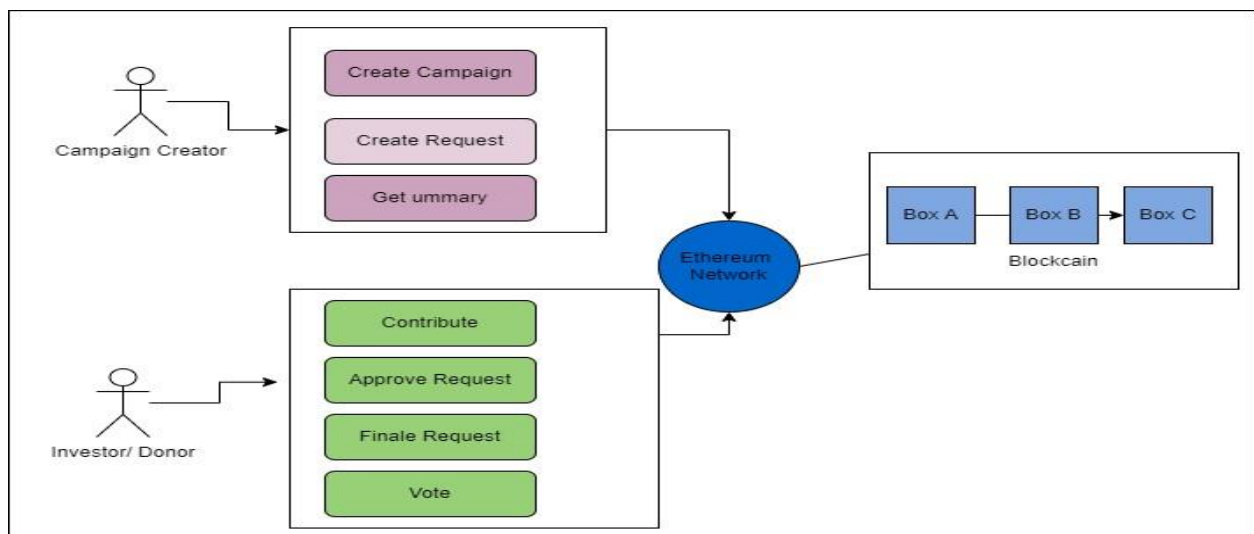


Figure 2: Working Model

[3].In a crowdfunding platforms that leverages the power of blockchain and the mechanism to bring together the most suitable for investors /donors and developers. This platform utilize the ethereum and solidity to create smart contracts, and it employs the proof of work (POW) consensus algorithm to ensure the safety and integrity of transitions. This platform is designed to serve to both organizations and individual users, but it does have the limitation of relying on the Rinkeby test network, which was discontinued on October 5, 2022. This platform provide unique path for investors/donors and developers to connect and collaborate on projects.

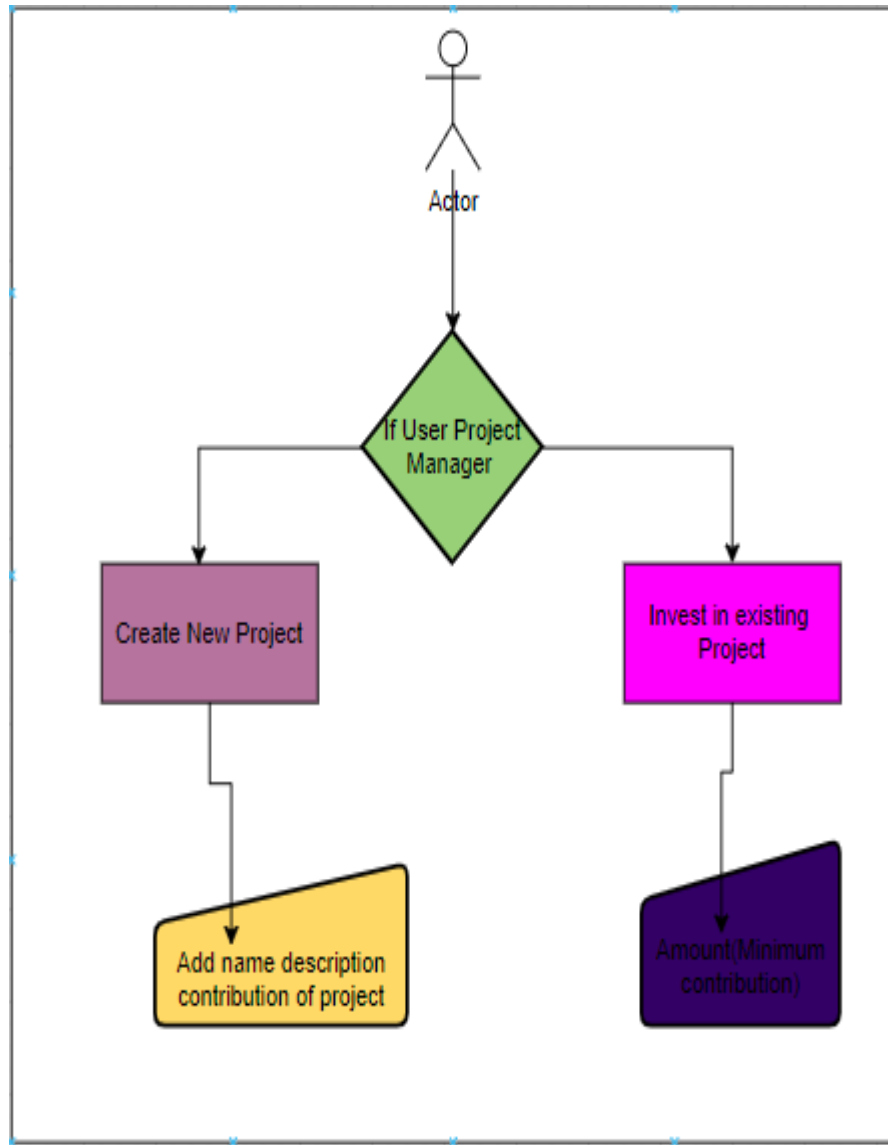


Figure 3: Creating or contributing to project

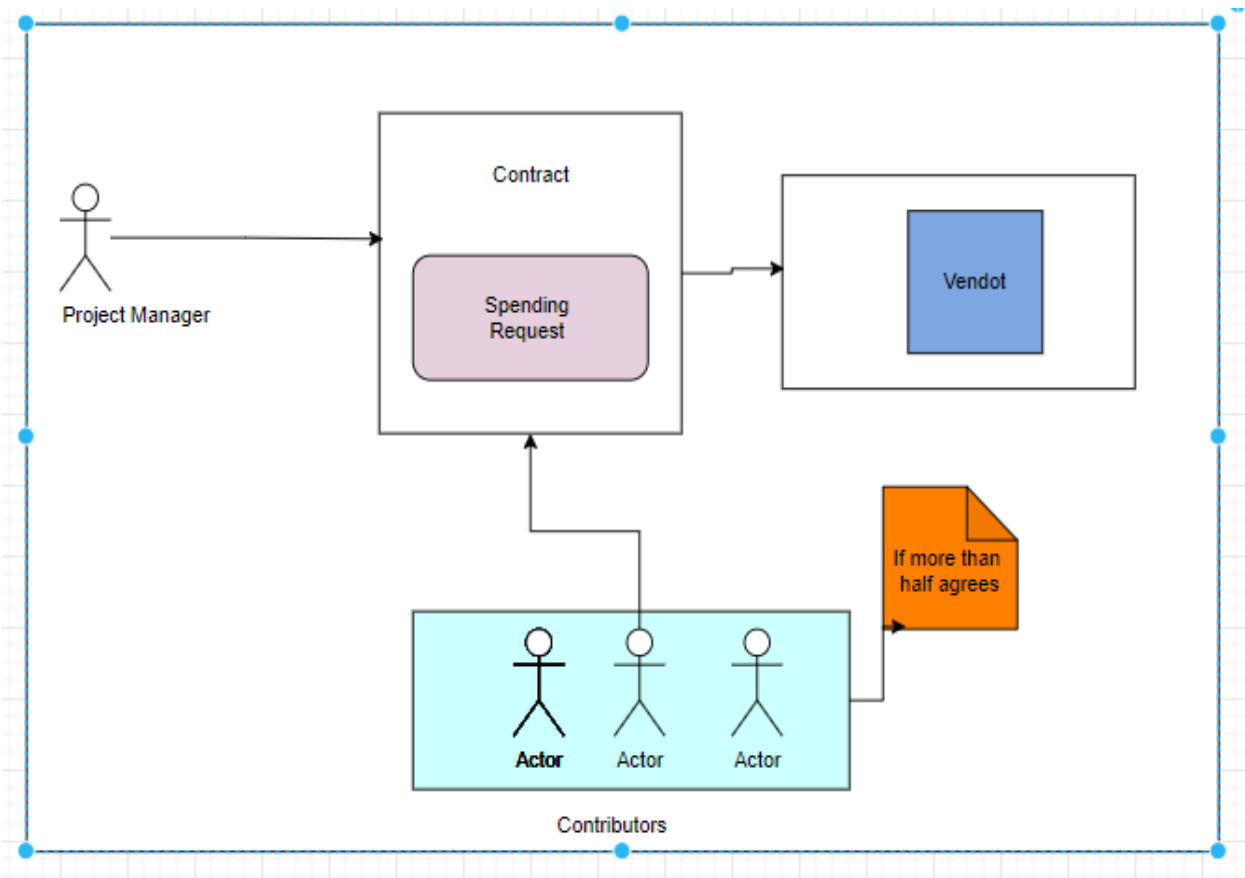


Figure 4: Voting system ensures money spent is in control of contributors

[4]. In a donation based crowdfunding platforms allows individual users and organization to raise money for charities business projects, and other causes. This platform utilizes solidity language to create smart contract and can be accessed through online social media or other platforms to reach a wide audience of potential sponsors, but this platform provides a convenient and effective path for users and organization to raise donations and bring their ideas to life.

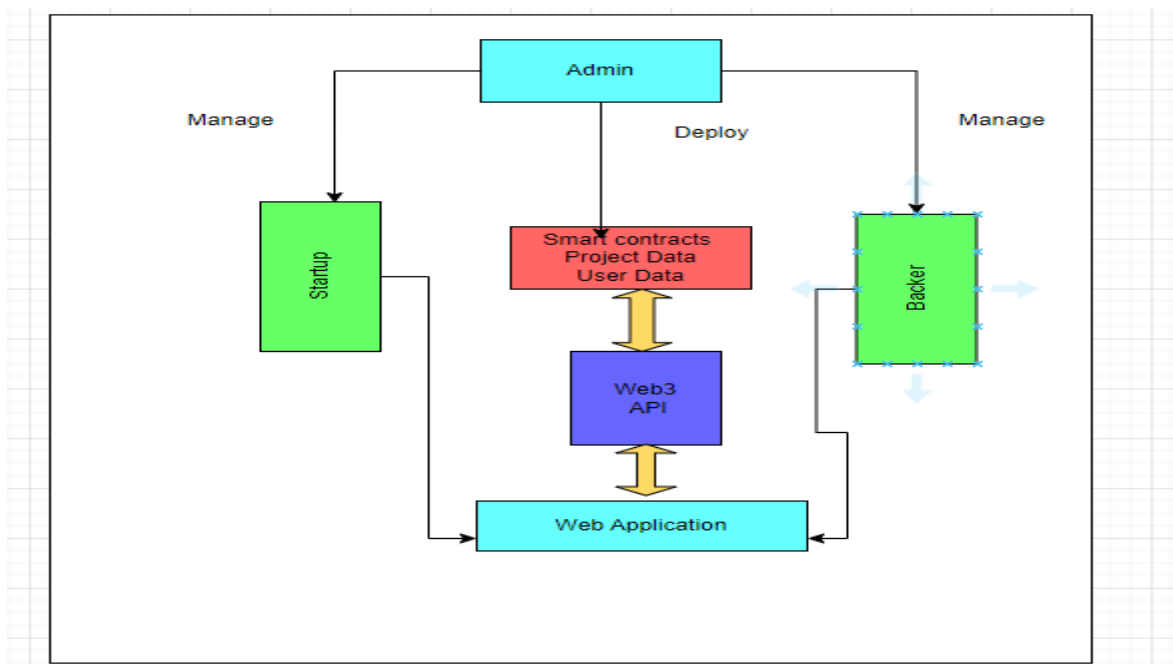
[5]. In a crowdfunding platform called WHIRL that is based on a pay-It-forward economic mode.

The model aims to create a positive feedback loop of generosity and giving, In which anyone who starts a campaign has already paid their dues by supporting other on this platform. WHIRL stand out from other crowdfunding platform called due to this rule as every other platform requires fund-raising to fulfill its obligations to backers. This platform utilizes the WHIRL method, which has demonstrated the ability to reach an optimal solution through multiple changes of bidding and perform better than other generic algorithms. But there are some limitations of this project including the requirement for users to collect karma points before launching a campaign and a lack of sufficient documentation. WHIRL provide unique path for individual users and organization to raise-funds and bring their ideas to life.

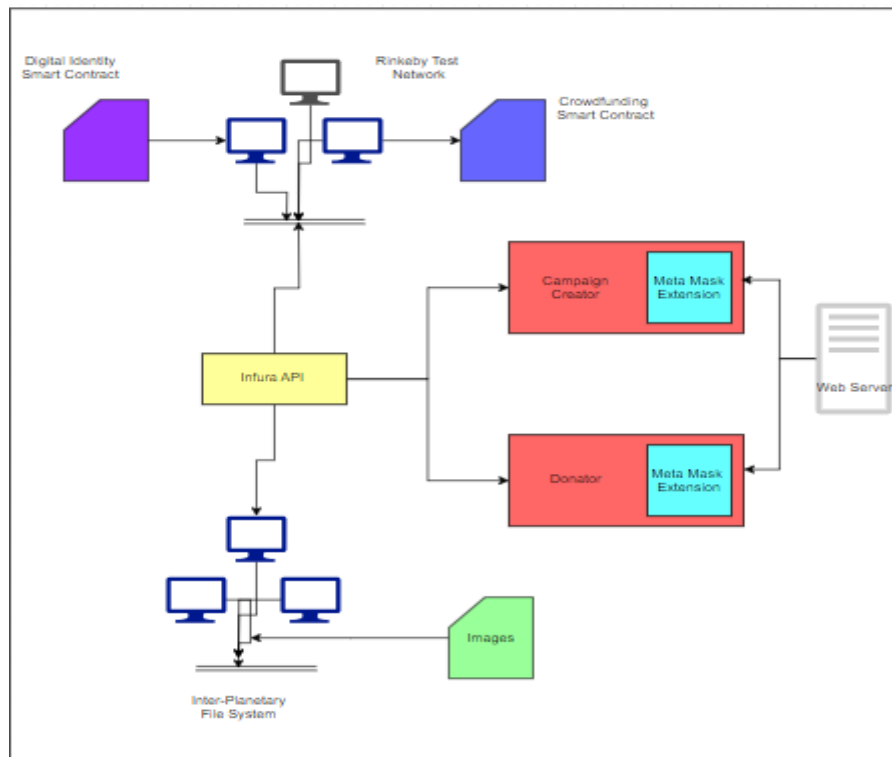
[6]. Bitfound is an online cryptocurrency platform. Developers and marketers can follow the trend of the network. Marketers can request the project and offer starting prices based on the cost, time and maintenance required. Developers may complete with different values at different rates to takes ownership of that project. Smart contract are used by developers and investors to reach the best solution for investors. The absence of a third party force proves fair to investors and developers alike as smart contracts or custom action phone numbers have not effect on bias.

Investor present their requirements and developers can complete on projects and they find the best deals that investors. Among them the ethereum network is used to send smart contracts between investors and developers for smart computing.

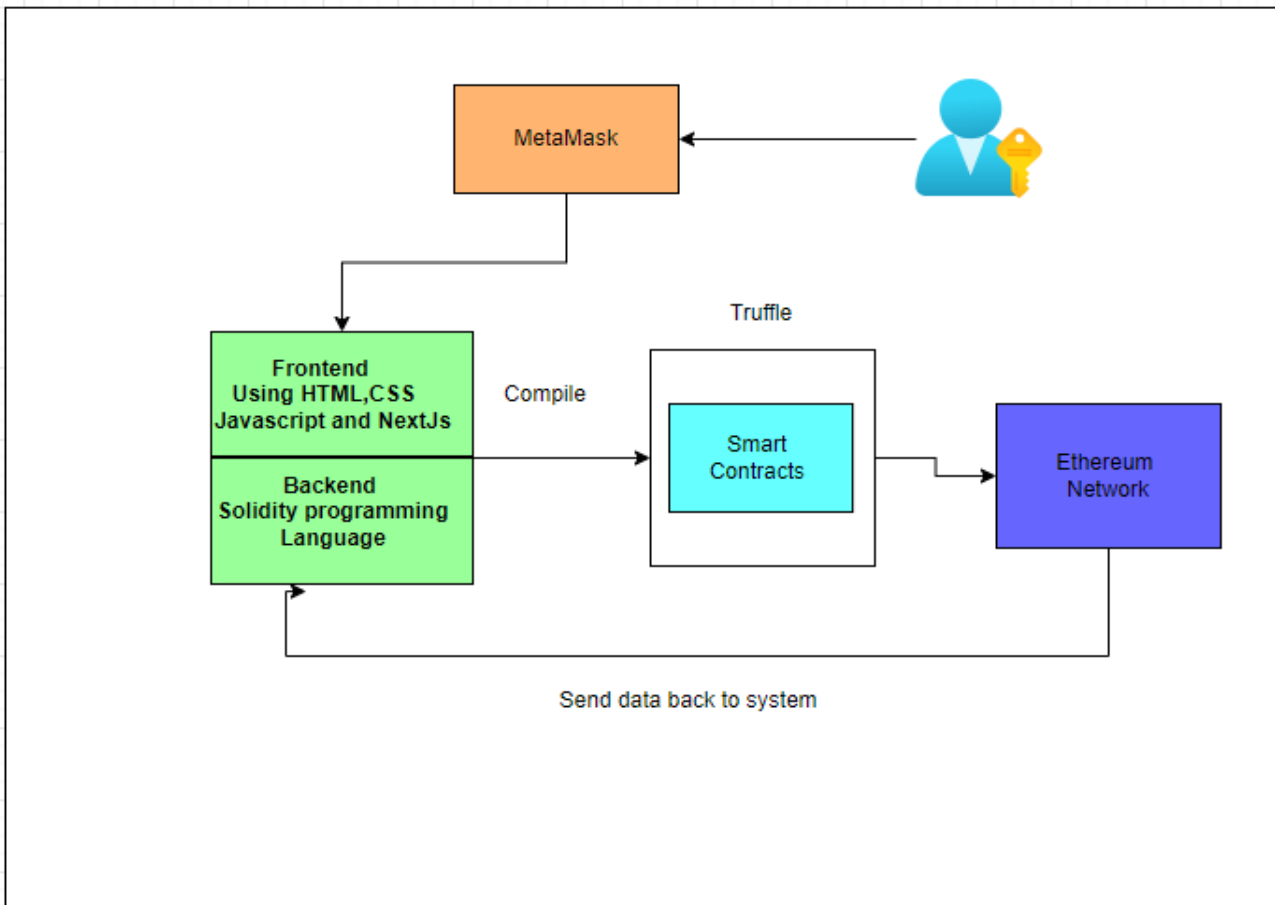
[7]. Crowdfunding application, unlike other application, only allow people to their money, but this platform also recognizes backers can track and whose returns will be guaranteed. This platform will also provide transparency between backers and startups, backers can track the progress of the various initiatives they have invested in. If the project does not continue, the money returned to the contributors/ sponsors. This app will be multiple user application with three different types of interface, Admin, Supporter and Startup. Administrators can agree to start a service. Startup can view the status of their operation and check the revenue over time. Supporters can see the progress of the projects and they can check the general information about other projects listed in the application.



Traditional Crowdfunding platforms [8]. Are vulnerable to data breaches, platform fees and high transaction and rampant fraud that occurs due to the anonymity of user identity. For example users cannot be identified when they commit cybercrime. The solution of digital identity management system using blockchain. The identity management system is an alternative to the know you customer(KYC) system. in KYC organization verify the identity of customers. The running cost of an organization using digital identity management is lower as compare to the cost of KYC. By implementing blockchain into digital identity management system we can track the malicious users and action taken against them. The blockchain is also helpful in digital identity management system to secures the system against centralized architecture attacks such as data breaches and DDOS.



3. Proposed System



The smart contracts for the Decentralize APP (DApp) is written in solidity to the personal Ethereum blockchain using Truffle. The frontend of the App is made using Nextjs and Web3.js is used to interact with Ethereum and smart contracts.

The system facilitates the campaign creation, contributing to campaigns and withdrawal within a blockchain based crowdfunding platform.

Following are the functionalities of D App.

3.1 Campaign creation

This system allows users to create crowdfunding campaigns. Users can provide details information about their project, including the funding purpose, description and timeline. This information is securely stored on the blockchain system ensuring immutability and transparency.

3.2 Contribution to campaign

Contributors can contribute funds to the campaigns of their choice. This system enables secure transactions by using blockchain technology which ensure the traceability integrity of contributors.

3.3 Smart contract Automation

Smart contracts, implemented using solidity programming language to automate various aspects of the contribution process. These smart contracts define the protocols and conditions for contributions, ensuring that funds are allocated correctly, transaction details are recorded on the blockchain.

3.4 Withdrawal process

When campaign is successfully funded, project administrators can initiate the withdrawal process to access the funds. This system verifies the authentication of the withdrawal request and based on the predefined rules and transfers the funds to the designated account or wallet securely. Withdrawal transaction are recorded on the blockchain for accountability.

3.5 Auditability and transparency

All transactions including contribution campaign creation and withdrawals are recorded on the blockchain, providing immutability and transparency. This system allows contributors and campaign creator to track the flow of funds verify the authentication of transactions ,that ensure accountability throughout the process.

4. Methodology

4.1. System Design and implementation

First we proposed overall system and design the structure of crowdfunding platform. We have needed different hardware and software resource for that purpose but we follow this outlines.

4.2. Requirement Analysis

Requirement analysis plays a vital role in the software development process. Without requirement analysis, we cannot deliver fully functional software. In this phase, we meet with stakeholders to collect requirements and ask questions. We then document the requirements properly. In requirement analysis, we deal with the overall flow of the system. In this process, we deal with the old crowdfunding system and how to integrate it with blockchain.

4.3 System Designing

After completing our front-end modules, we can try to integrate with the blockchain using Web3.js, which allows interaction with the Ethereum network. For integration that involves the blockchain using MetaMask, a secure wallet and access to the Ethereum network are provided. MetaMask allows users to contribute funds, interact with smart contracts, and track all transactions done on the blockchain.

4.4 Integration with blockchain

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4.5 Smart Contract Development

Smart contracts are developed using the Solidity programming language, specifically designed for Ethereum. Smart contracts can mention fund-raising campaigns, handle contribution logic, and automate reward distribution and withdrawal processes. The Truffle framework is used to compile and deploy smart contracts to the blockchain. Ganache is a local blockchain database used to test and debug smart contracts during the development environment.

4.6 Deployment

Once the testing phase is complete, our crowdfunding system will be ready for deployment. We will first deploy the smart contracts to the Ethereum network, and then deploy the frontend to Vercel, which is an excellent platform for deploying Next.js applications. After that, users will be able to easily access the crowdfunding platform through their web browser, connect their MetaMask wallets, and begin contributing funds or creating campaigns.

5. Result and Analysis

Based on the implemented crowdfunding system, the following are the observations and results.

5.1 Creation of Efficient Campaign

The system efficiently facilitates the creation of campaigns, allowing users to easily set up their fundraising projects. System users can input details such as the funding purpose, project description, and timeline. After that, the system ensures that the campaign is created through a lightweight process.

5.2 Secure Contribution Management

The system provides secure management of contributions by participants, recording all contributions on the blockchain network to ensure immutability and transparency. Smart contracts automatically handle the contribution process, verifying the authentication of transactions and efficiently allocating funds to the respective campaigns.

5.3 Transparent Fund Tracking

This platform offers transparency in tracking the flow of funds within campaigns. System participants can view detailed information about the funds raised, including the total amount contributed, individual contributions to campaigns, and the funding status of each campaign. This mechanism builds trust among contributors and allows them to easily verify financial transactions.

5.4 Enhanced the Security

Using blockchain technology, the crowdfunding system enhances privacy and security. All participant information and financial transactions are securely encrypted and permanently stored on the blockchain, protecting them from unauthorized access. The decentralized nature of blockchain significantly reduces the risk of data breaches and increases the overall security of the system.

5.5 Immutable Records

Blockchain is an immutable technology that provides immutable and auditable records of all crowdfunding transactions and activities. Every transaction, contribution creation, and reward distribution is permanently recorded on the blockchain, allowing for easy auditing and dispute resolution if necessary.

5.6 Potential for Scalability

The proposed blockchain crowdfunding system is scalable, flexible, and efficient. It can manage a growing number of campaigns and contributions, and incorporate additional features such as social gathering campaign analysis or integration with other networks.

The implemented system has demonstrated its efficiency in campaign creation, secure and reliable contribution distribution, and empowering contributors and project creators during their fundraising programs.

6. Challenges

Following are the challenges for implementing blockchain system.

6.1 Scalability

Blockchain networks such as Ethereum may face scalability challenges, resulting in slower transaction processing and higher fees during peak periods. These challenges can make it difficult to accommodate a large number of crowdfunding participants and campaigns simultaneously.

6.2 User Onboarding and Awareness

So blockchain is new technology and many users unfamiliar with blockchain. Many people are not aware blockchain and its Application Lack of awareness it is a big barrier and We need to give awareness for people and arrange the workshop for user to adopt the system This system ensuring a smooth onboarding experience and provide user friendly interface for user and that is helpful for users of crowdfunding platforms.

6.3 Regulatory uncertainty

Currently there is no clear regulatory system for blockchain crowdfunding in Pakistan. This uncertainty is difficult for people to use the blockchain crowdfunding platforms.

6.4 Availability of Resources

There are many difficulties can occur during maintenance of this system because there are limited resources are available solidity is a new programming language as compare to c, c++ and java. So limited documentation and resources are available.

7. Conclusion

The traditional crowdfunding success rate in Pakistan is 50% if we implement blockchain based crowdfunding then increase the trust of donors reduce the cost of transactions and we can tracked transactions to increase the transparency of system.

However, there are many challenges that need to be addressed is label to implemented blockchain based crowdfunding. Including scalability, regulatory uncertainty, awareness and limited resources.

The government of Pakistan and their institution of Pakistan can play the vital role in promoting the blockchain based crowdfunding and implementation of this system on different startups to raising the funds.

This system is helpful to increase the transparency, efficiency and secure the transactions. Another benefit of blockchain is used in crowdfunding the data is permanently recorded and immutable.

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